

From Potholes to Playoffs: Paving the Way for the 2026 NFL Draft in Pittsburgh with Predictive Modeling

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1. Executive Summary

In preparation for the 2026 NFL Draft, which is expected to bring hundreds of thousands of visitors and major national attention to Pittsburgh, our project proposes a data-driven solution to reduce 311 infrastructure complaints, with a specific focus on pothole-related service requests to proactively improve the city's road conditions. In order to assist the Department of Public Works (DPW), we developed a set of machine learning models aimed at transforming Pittsburgh's complaint response system from reactive to proactive, enabling smarter resource allocation and long-term infrastructure savings.

Our solution consists of three integrated components. First, we applied geospatial clustering to identify persistent complaint hotspots across neighborhoods and complaint types. Second, we built a binary classification model to flag "repeat complaint zones" for areas that consistently receive call requests for the same issue. Finally, we developed a time-series forecasting model to predict departmental workload by week, with an emphasis on pothole service volumes, enabling preemptive staffing and budget planning. Together, these tools empower the DPW to prioritize maintenance, reduce complaints, and enhance the visitor and resident experience ahead of this high-impact event.

2. Background and Motivation

Pittsburgh is set to host the 2026 NFL Draft, an event expected to draw hundreds of thousands of visitors and considerable national attention. The 2024 draft in Detroit highlighted the scale of such events, attracting over 775,000 attendees and generating \$213.6 million in economic activity (Detroit Sports Commission, 2024). To ensure a positive experience for visitors and residents alike, the city of Pittsburgh must prioritize highly visible aspects of infrastructure, especially road conditions.

One of the most persistent and visible public complaints in Pittsburgh is pothole damage. Potholes are not just a soresight; they pose safety risks, damage vehicles, and signal underlying infrastructure issues. In 2023 alone, Pittsburgh residents filed over 15,000 pothole-related 311 complaints (City of Pittsburgh, n.d.-a). Pothole formation is exacerbated by freeze-thaw cycles and heavy traffic, both of which are common in the region (McConnell & Associates, n.d.). Without timely intervention, minor pavement issues can escalate into major roadway failures, leading to costly repairs and safety hazards.

Currently, the city's response to pothole issues is predominantly reactive, relying on residents to report damage through 311 requests (City of Pittsburgh, n.d.-b). This approach delays response times, overlooks emerging patterns, and strains maintenance resources. Studies have shown that preventive road maintenance is 6 to 10 times more cost-effective than reactive repair (Minnesota Local Road Research Board, 2000). Moreover, undetected repeat complaint zones may indicate chronic maintenance failures or road segments in need of structural upgrades.

To address these challenges, the DPW must shift toward a more proactive strategy. We aim to provide a strategy that leverages data-driven tools to anticipate repeat pothole complaints, prioritize high-risk areas, and streamline resource deployment. Doing so would not only improve road safety and resident satisfaction but also present Pittsburgh as a city that is clean, efficient, and prepared for the national

spotlight. With the 2026 NFL Draft fast approaching, the DPW has a limited window to act. Using predictive tools like ours can help ensure that Pittsburgh presents itself as a well-maintained and prepared host city, especially when time and resources are limited.

3. Related Work

Most U.S. cities, including Pittsburgh, rely on complaint-driven 311 systems to manage infrastructure issues such as potholes, snow removal, and waste collection (City of Pittsburgh, n.d.-a). While effective for addressing individual reports, these systems are reactive by design and offer limited ability to detect chronic and systemic issues, particularly in the case of potholes which often signal underlying roadway deterioration. Prior research has explored predictive modeling in pavement management, using historical road condition data to guide cost-effective interventions and prioritize resurfacing projects (Minnesota Local Road Research Board, 2000; City of Davis, n.d.).

Some municipalities have enhanced service response through the use of weather and traffic data, especially for snow removal, but few have implemented integrated machine learning pipelines aimed specifically at reducing pothole-related complaints. Our work builds on these efforts by combining geospatial clustering, repeat pothole complaint prediction, and time-series forecasting to support preventive maintenance. This cross-functional approach contributes a novel, scalable framework for municipal planning that aligns pothole response strategies with operational readiness.

4. Problem Formulation and Project Overview

The goal of this project is to reduce the volume of 311 service complaints in Pittsburgh—with a primary focus on pothole-related requests—by developing predictive tools that support proactive infrastructure maintenance. We frame this as a set of machine learning tasks designed to identify high-risk areas, anticipate recurring issues, and inform resource planning.

Our solution includes three components:

1. **Geospatial Complaint Hotspot Clustering:** Applies unsupervised learning to uncover persistent, high-density clusters of *pothole* complaints, revealing geospatial patterns across Pittsburgh neighborhoods.
2. **Repeat Complaint Zone Prediction:** Applies binary classification to flag if a location is likely to have a repeated pothole related complaints.
3. **Departmental Workload Forecasting:** Uses a time-series forecasting model to predict weekly pothole-related complaint volumes by department and neighborhood for improved staffing and scheduling.

These components collectively serve as a decision-support framework for the Department of Public Works to prioritize interventions, reduce service disruptions, and allocate resources more efficiently in the lead-up to the 2026 NFL Draft and beyond.

5. Data Description and Exploratory Analysis

5.1 Method Used

The initial phase involved conducting exploratory data analysis (EDA) on the 311 service request dataset to understand its structure, identify potential issues, and guide further preprocessing. Key steps included viewing the first and last few rows to gain a general understanding of the entries, followed by inspecting dataset metadata using `.info()` to assess column types. Summary statistics were generated using `.describe()` to evaluate central tendencies and detect potential outliers. The presence of missing data was examined both in count and percentage form, and duplicate entries were quantified. Furthermore, the uniqueness of each column was assessed to differentiate between categorical and continuous features, and a correlation matrix was computed to understand relationships among numerical variables.

Subsequently, data cleaning involved dropping redundant columns such as `census_tract` and `cross_street`. A subset of relevant features was selected to form a new DataFrame focused on machine learning use cases. The dataset was then filtered to include only records related to "Potholes" in the `request_type_name` column, aligning with the intended scope of analysis. Missing data was reevaluated in this filtered subset, and any records containing null values were removed to ensure data integrity. The resulting cleaned dataset provided a robust foundation for downstream geospatial clustering and modeling tasks.

5.2 Findings and Important Insights

The raw 311 Data has 815417 rows and 29 columns. The following columns have a relatively higher percentage of null values: `cross_street` (~87%), `street_id` (~38%) and `cross_street_id` (~38%). However, the missing values are considerably higher for other request types as compared to Potholes. Secondly, since our analysis only focuses on potholes, we drop irrelevant columns like `census_tract` and `cross_street`. We filter out the data frame for only request types related to potholes and are left with the following columns: `'group_id'`, `'num_requests'`, `'status_code'`, `'dept'`, `'request_type_name'`, `'street'`, `'cross_street_id'`, `'neighborhood'`, `'police_zone'`, `'council_district'`, `'geo_accuracy'`, `'latitude'`, `'longitude'`, `'create_date_et'`. This resultant data frame has 67502 rows and 14 columns. We recheck for null values and find that the null values make up less than 0.5% of the dataset. Then, we dropped all rows which had null values associated with them using the `dropna` function. The final DataFrame has 66248 rows and 14 columns.

6. Component 1: Geospatial Complaint Hotspot Clustering

6.1. Objective and Approach

The objective of this machine learning model is to uncover spatial patterns in Pittsburgh's pothole-related complaints (using 311 data) and identify actionable clusters of high-volume problem areas. Using geospatial and temporal data from thousands of service requests, the goal is to support data-driven street

maintenance planning by revealing not only where complaints are concentrated, but how they naturally group together across the city. The approach involved geospatial preprocessing, clustering with K-Means, and visualizing complaint density using interactive heatmaps.

6.2. Modeling Techniques and Implementation

The model pipeline began with filtering only high-quality data points (`geo_accuracy == 'EXACT'`) and parsing complaint timestamps. Seasonal context was introduced by extracting the month and encoding seasons using one-hot encoding. Additionally, `request_type_name` was frequency-encoded to quantify complaint categories. A final feature matrix was constructed including latitude, longitude, number of requests, complaint type, and seasonal indicators.

The data was scaled using `StandardScaler`, and K-Means clustering (sklearn used) was applied to group complaints into four clusters ($k = 4$), chosen based on the Elbow Method. Post-clustering, the model aggregated complaints by street and normalized their intensity using `MinMax` scaling. The top 300 complaint-heavy streets per cluster were retained for visualization, reducing noise and clutter in the analysis.

For comparative analysis, the `folium` library was used to create heatmaps. Map A displays a basic complaint density heatmap (filtered for locations with ≥ 5 complaints), Map B highlights the top 50 streets by complaint volume, and most importantly Map C integrates clustering with complaint volume using layered heatmaps.

6.3. Results and Evaluation

The clustering yielded four spatially coherent groups of complaints across Pittsburgh. The Silhouette Score was approximately 0.51, indicating moderate clustering quality. This means that the clusters are reasonably well-separated but show some overlap, which is typical in geospatial data. Each cluster contained distinct complaint-dense zones, suggesting meaningful separation between urban subregions.

Map A visualized complaint hotspots using raw data, while Map B spotlighted specific streets with the highest complaint counts. Map C revealed broader spatial structures, where clusters aligned with known neighborhoods such as East Liberty, South Side, and parts of Wilkinsburg, confirming the model's spatial relevance.

6.4. Interpretation and Insights

The analysis shows that a small number of locations account for a disproportionate volume of pothole complaints. Specifically, Map B highlighted that the top 50 streets alone represent a substantial share of total complaints. Map A confirmed that complaint activity is not randomly distributed but concentrated in repeat-problem areas.

Map A showed that complaints are concentrated in areas like Downtown, East Liberty, or South Side. On the other hand, Map B delivers targeted insight into the worst-performing streets in terms of maintenance

issues. The top-reported streets include: Liberty Ave with over 200 complaints, Penn Ave and Carson St with 150–180 complaints each.

Most importantly, Map C uncovered four actionable geographic clusters of complaints: (a) Low complaints, (b) Moderate Complaints, (c) High Complaints and (d) Very High Complaints. These clusters are not only dense in volume but also exhibit internal spatial cohesion, making them ideal targets for location-based maintenance efforts. The accompanying heatmap legend helped interpret the severity of complaint intensity, ranging from blue (low) to red (very high). This provides city planners a clear and intuitive tool for identifying priority areas, moving from a reactive to a proactive infrastructure strategy.

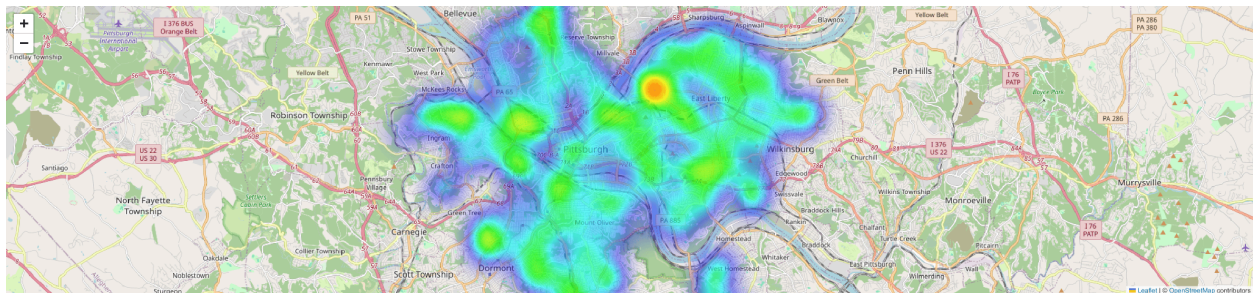
6.5. Maps: Visualization & Interpretation

(i) **Map A** visualizes a heatmap of pothole complaints across Pittsburgh. It begins by aggregating complaints at each geographic coordinate, counting the number of unique complaint groups (group_id). To avoid clutter from insignificant reports, it filters out locations with fewer than 5 complaints, ensuring that only meaningful hotspots are displayed. The complaint counts are then log-scaled and normalized, allowing both low and high-frequency areas to be visible without being dominated by outliers.

With an enlarged radius and blur, the heatmap highlights broad urban zones where pothole problems are most concentrated. These may include areas like Downtown, East Liberty, or South Side.

This map is highly effective for macro-level infrastructure planning, revealing where the greatest community dissatisfaction is concentrated over time. It's ideal for strategic planning or hotspot detection.

Note: Areas with higher complaint density are coloured in orange, areas with moderate complaint density are coloured in yellow, and areas with lower complaint density are coloured in green.

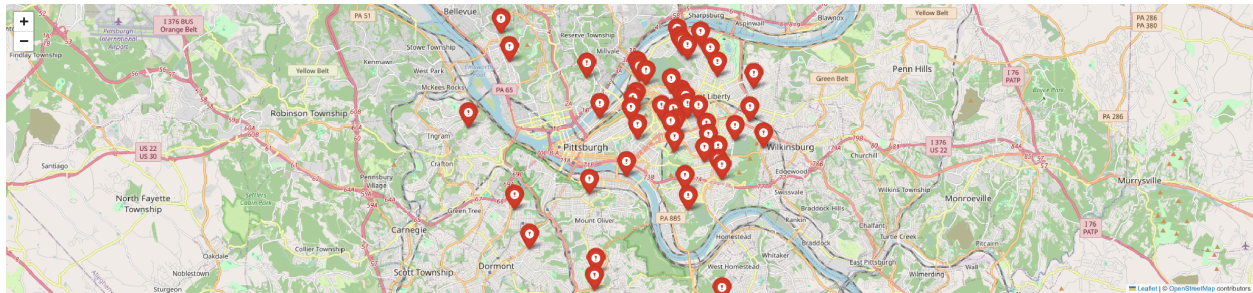


(ii) **Map B** complements Map A by shifting focus from broad geographic trends to specific street-level accountability. It aggregates complaints by street and identifies the top 50 streets with the highest number of complaints. Each of these streets is then represented on a map with a labeled marker, showing both the street name and the exact number of complaints received.

This map is particularly useful for decision-makers and field teams, as it clearly identifies which streets have the greatest need for inspection or repair. The use of red marker icons and tooltips ensures that these problematic streets stand out and are easy to locate. Unlike Map A, which shows patterns, Map B delivers

targeted insight into the worst-performing streets in terms of maintenance issues. The top-reported streets include: Liberty Ave with over 200 complaints, Penn Ave and Carson St with 150–180 complaints each.

Note: Using the code/jupyter notebook, you can hover on the red markers to get more information about the street like its name, number of complaints and more.

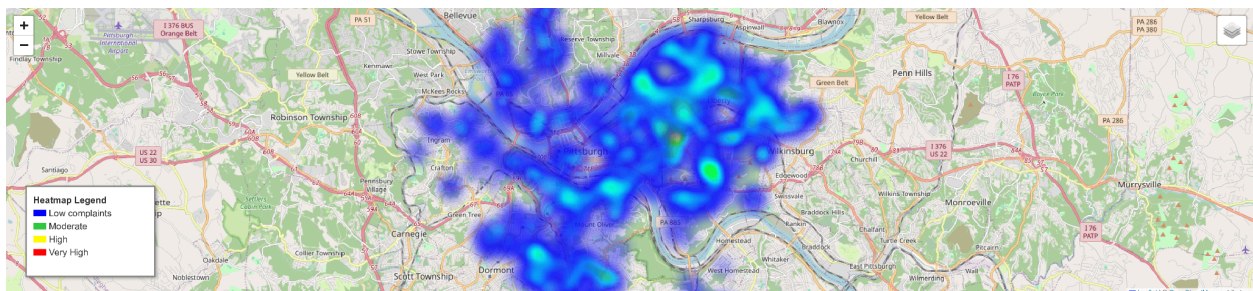


(iii) **Map C** visualizes the results of applying K-Means clustering ($k = 4$) to Pittsburgh's pothole complaint data. Each complaint was grouped based on location and potentially seasonal or volume-based features. Within each cluster, the map highlights the top 300 complaint-heavy streets, using a heatmap to represent intensity. The clustering reveals underlying spatial groupings. For instance, you might see that certain neighborhoods consistently fall into one cluster, suggesting shared infrastructure or traffic patterns.

Importantly, this version of the map includes a default Folium color gradient (blue to red), with a custom legend explaining the color scale. Blue areas indicate streets with fewer complaints, while red areas show very high complaint density. Although the map does not display raw points or individual street labels, it presents a clean and interpretable picture of where pothole issues are concentrated within spatial clusters. Each cluster is toggleable, allowing users to focus on one region at a time.

This map is especially useful for identifying broader zones of concern, such as neighborhoods or districts where infrastructure repairs could be prioritized strategically across multiple streets, rather than reacting to individual complaints. The accompanying legend helps users interpret color intensities without needing to dive into the raw numbers.

Note: The legend on the map figure shows low complaints cluster in blue, moderate in green, high in yellow and very high in red.



7. Component 2: Repeat Complaint Zone Modeling for Focused Intervention

7.1. Objective and Approach

The goal of this component was to identify geographic areas, identified by `cross_street_id`, that are prone to receiving repeated 311 service requests for potholes. By predicting whether a new complaint is part of a recurring pattern at a given location, the city can proactively allocate resources or escalate unresolved cases. We framed this as a binary classification task: predicting whether a given complaint was part of a cluster of repeat complaints in the previous 30 days. A request was considered to be a repeat request if two other pothole requests at the same cross street were made in the previous 30 days, making the current request the third request in 30 days.

7.2. Modeling Techniques and Implementation

We used a logistic regression model trained on structured 311 complaint data, incorporating features such as `cross_street_id`, `request_type_name`, and `'create_date_et'`. Through feature engineering, we created two features to predict repeat complaints:

'num_repeat_complaints': Counts the number of times that the same request type had occurred at the same location in the past 30 days.

'multiple_repeats': A binary feature that assigns a value of 1 to a repeat if two other repeats of the same type and cross street id were made in the previous 30 days.

Four different models were run to predict if a pothole request at certain cross street ids had a `multiple_repeats` value of 1: Logistic Regression, Random Forests, Gradient Boosting, and XGBoost. Since there was a class imbalance between repeat and non-repeat complaints, the same four models were also run with and without applying the SMOTE method on the training data. The model pipeline used one hot encoding for categorical variables. Hyperparameters were tuned using a randomized grid search with cross-validation.

7.3. Results and Evaluation

Figure 7.3: Model Results with and without SMOTE

Model	Precision	Recall	F1 Score	Accuracy
Logistic Regression	0.46	0.78	0.58	0.79
Logistic	0.46	0.78	0.58	0.79

Regression with SMOTE				
Random Forest	0.5	0.71	0.59	0.82
Random Forest with SMOTE	0.47	0.78	0.58	0.80
Gradient Boosting	0.69	0.38	0.49	0.85
Gradient Boosting with SMOTE	0.55	0.67	0.60	0.84
XGBoost	0.73	0.31	0.43	0.85
XGBoost with SMOTE	0.56	0.66	0.60	0.84

Since the goal of this component was to identify locations with a high likelihood of repeated pothole complaints, recall was the metric that was prioritized. A higher recall value means that more areas that have repeat pothole requests are being identified as repeat zones. While this meant selecting a model with lower precision, we believe it is in the city of Pittsburgh's best interests to falsely identify an area as a repeat request rather than not identify actual repeat request areas. The models that achieved the highest recall was Logistic Regression with and without SMOTE, with a recall of 0.78. This indicates that the models correctly identified 78% of all true repeat complaints in the validation set, making it the most effective at surfacing zones of potential recurring service issues.

Interestingly, applying SMOTE did not improve recall for logistic regression, likely because the model was already effective at accounting for class imbalance by setting `class_weight = 'balanced'`. However, SMOTE did improve performance for the Gradient Boosting and XGBoost models.

Ultimately, we selected the logistic regression model without SMOTE for deployment due to its high recall and overall high performance across validation metrics. Since the Logistic Regression delivered the same results with and without SMOTE applied, we do not believe it is necessary to apply SMOTE to this model. While some models had higher accuracy and precision, their lower recall made them less suitable for this task that prioritizes catching as many pothole repeat complaint areas as possible.

7.4. Interpretation and Insights

The repeat complaint zone model offers a practical method for identifying areas with recurring pothole reports. Its high recall enables the city to proactively address locations that are likely to receive repeated complaints, helping prevent service delays and resident frustration. While precision was moderate, the operational benefit of identifying true repeat zones outweighs the cost of occasional false positives. Targeting repeat request areas can improve public safety, as these locations likely involve more severe or

persistent potholes that pose greater risks to drivers and pedestrians. This approach allows the DPW to flag high-priority locations for early intervention, improving response efficiency and potentially reducing total repair volume over time.

8. Component 3: Forecasting Departmental Workload for Resource Allocation

8.1. Objective and Approach

This component aims to support the Department of Public Works in planning ahead for pothole-related service requests by forecasting weekly complaint volumes. Accurate short-term forecasts allow the DPW to schedule crews, allocate equipment, and manage budgets more effectively. We framed this as a supervised time-series regression task using historical 311 request volumes. The target variable was the number of pothole complaints in a given neighborhood, one week ahead.

8.2. Time-Series Modeling and Implementation

We developed and refined a series of XGBoost models across three iterations, each improving on feature selection and time representation:

- **Time Features:** Month, Week, and Season were extracted from timestamps and later transformed into cyclical encodings to reflect the periodic nature of calendar time.
- **Lag Features:** Each model included 4 prior weeks of complaint volume to capture short-term temporal dependencies.
- **Smoothing:** A 4-week moving average was introduced to reduce noise and improve signal stability.
- **Scope:** The final model was filtered to pothole-related requests and generated forecasts at the neighborhood level (e.g., Shadyside, Squirrel Hill South).

Weekly aggregation and test/train splits were handled chronologically to preserve time-order integrity.

8.3. Results and Evaluation

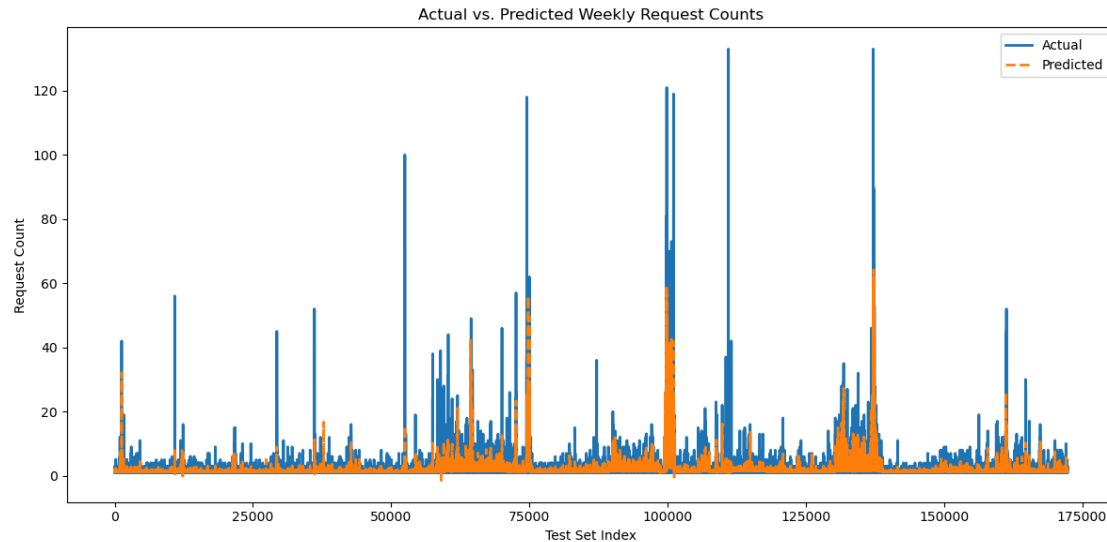
We evaluated each iteration using mean squared error (MSE), root mean squared error (RMSE), mean absolute error (MAE), and R^2 . Each model was validated using a held-out set of recent weeks and plotted against actual historical values.

Version 3.01 – Baseline XGBoost with Basic Time and Lag Features

- **Features:** Raw Month, Week, Season + 4 lags
- **Performance:**
 - MSE: 5.0782

- RMSE: 2.2535
- MAE: 0.8122
- R^2 : 0.3713
- Summary: The model identified general trends but lacked precision, likely due to unprocessed time variables and the absence of smoothing.

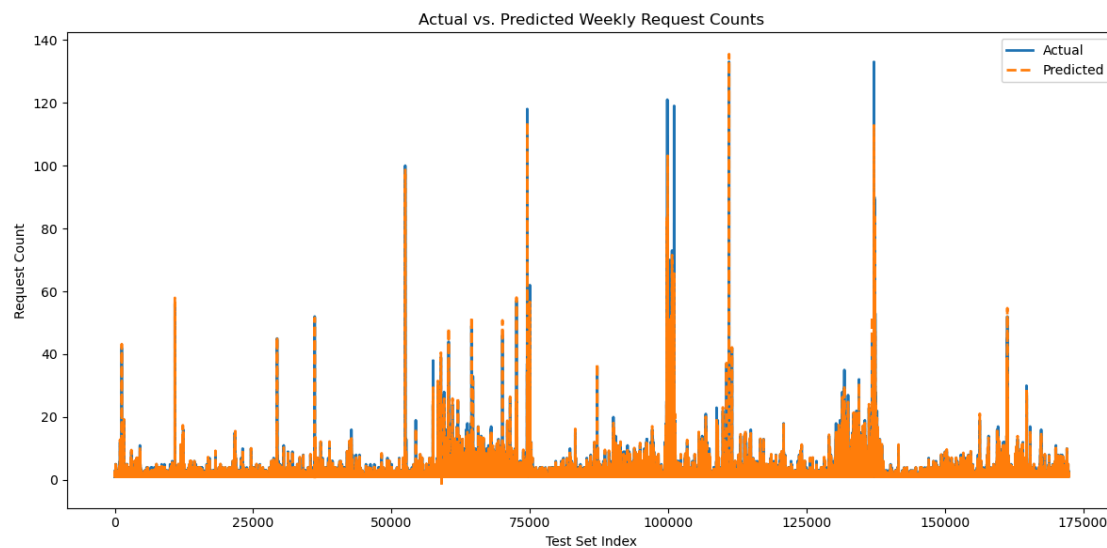
Figure 8.1: Predicted vs. Actual Weekly Request Volume – v3.01



Version 3.02 – Cyclical Time Features + 4-Week Moving Average

- Enhancements:
 - Cyclical encoding of Month, Week, Season
 - 4-week moving average added as a feature
- Performance:
 - MSE: 0.5565
 - RMSE: 0.7460
 - MAE: 0.1355
 - R^2 : 0.9311
- Summary: Accuracy improved significantly, especially for capturing local week-to-week variation in complaint counts.

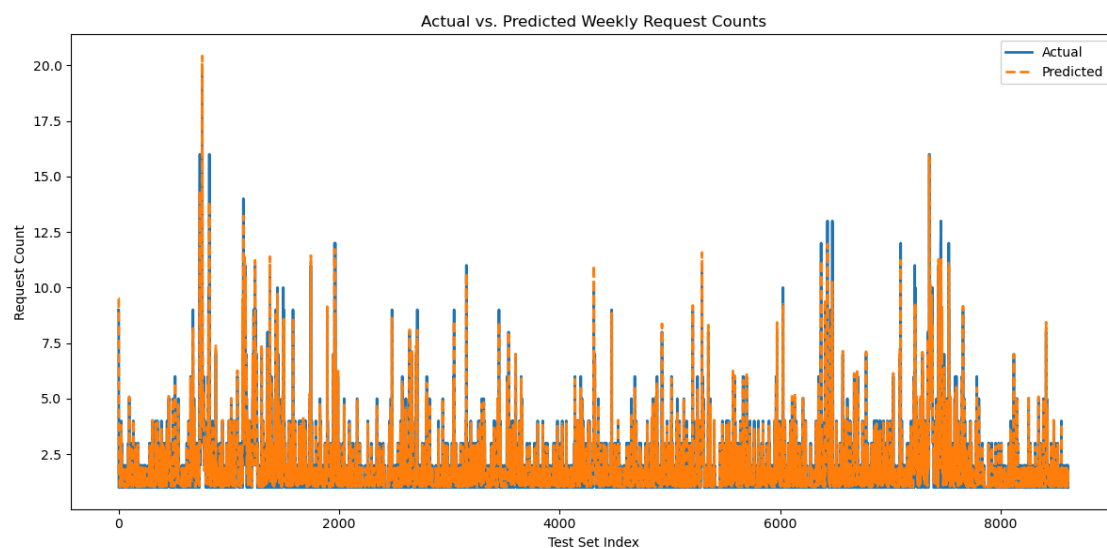
Figure 8.2: Predicted vs. Actual Weekly Request Volume – v3.02



Version 3.03 – Final Model: Pothole-Only Forecasting by Neighborhood

- Refinements:
 - Filtered to pothole requests only
- Performance:
 - MSE: 0.0698
 - RMSE: 0.2642
 - MAE: 0.1566
 - R^2 : 0.9768
- Summary: The final model achieved strong predictive accuracy and is well-suited for use in real-world DPW resource allocation.

Figure 8.3: Predicted vs. Actual Weekly Request Volume – v3.03



8.4. Interpretation and Insights

The forecasts aligned with known seasonal trends: pothole complaints increased in colder months due to freeze-thaw cycles and road wear. The final model's high accuracy supports its use in operational planning. The DPW can anticipate week-level workload spikes, pre-deploy road crews, and minimize overtime costs. Moreover, predictions at the neighborhood level allow for more equitable distribution of resources and attention. This component demonstrates that even with limited features, machine learning models can deliver actionable insights for municipal planning when well-tuned and properly scoped.

9. Bias and Fairness Audit

While we did not conduct a formal bias or fairness audit as part of this project, we recognize that predictive models used in public service contexts must be carefully evaluated for equitable outcomes. In the case of 311 complaint modeling, geographic and socioeconomic disparities could influence both complaint volumes and how resources are allocated in response. For example, residents in higher-income neighborhoods may be more likely to report issues, potentially skewing the training data toward areas with greater civic engagement rather than greater need.

To ensure that proactive maintenance strategies do not reinforce existing inequities, future work should include disaggregated model evaluations by neighborhood, council district, or census tract. Classification performance in the repeat complaint model, for instance, could be audited for false positive or false negative rates across different areas. Similarly, workload forecasts should be assessed to ensure that resource recommendations do not systematically under-serve communities with historically lower complaint rates but persistent infrastructure challenges.

In future deployments, incorporating fairness-aware metrics and validating predictions with on-the-ground conditions could help ensure that model-driven decisions promote both efficiency and equity.

10. Policy Recommendations

To maximize the impact of our predictive tools ahead of the 2026 NFL Draft, we recommend that the Department of Public Works adopts a more proactive approach to pothole management based on our models.

By using the geospatial clustering model, the DPW can identify persistent complaint hotspots and prioritize these areas for targeted maintenance. This allows the city to go beyond individual complaints and instead address underlying infrastructure patterns, improving both efficiency and public safety. Integrating these spatial insights into existing operations could help the DPW allocate resources more strategically and demonstrate a commitment to visible improvements before the national event.

The repeat complaint zone model provides an opportunity to shift from reactive to preventive service delivery. By flagging `cross_street_ids` that are likely to experience recurring pothole issues, the city can preemptively schedule inspections or repairs in these high-risk locations. While this approach may result in occasional false positives, it ultimately supports earlier intervention and minimizes long-term damage.

The DPW should incorporate these predictions into their internal scheduling systems and treat high-risk zones as early-warning indicators for deteriorating road conditions.

The time-series forecasting model supports forward-looking planning by predicting weekly pothole complaint volumes by neighborhood. The DPW can use these forecasts to adjust staff schedules, prepare materials, and manage seasonal demand more effectively. This model is especially important during the winter months when complaint volumes historically surge. By embedding forecasts into routine planning, the city can reduce costly emergency responses and maintain smoother operations throughout the year.

11. Limitations and Future Work

While our models demonstrate promising results, several limitations should be considered when interpreting their outcomes or deploying them at scale. The foundation of our work is 311 complaint data, which reflects resident-reported issues. This data may contain reporting biases, as some neighborhoods, possibly higher-income or more engaged communities, are more likely to file service requests than others. As a result, the models may have underrepresented pothole activity in areas with lower civic engagement or areas that have inappropriate access to cell phone service to make a 311 call.

One key limitation of our current models is the narrow feature set available from the 311 complaint dataset. Important variables that could enable a more comprehensive and accurate analysis, such as pavement condition ratings, historical maintenance schedules, traffic volumes, or road safety data, were not available and therefore not included in model development. These data sources are likely to be highly correlated with pothole formation and recurrence, and their integration could potentially enhance predictive performance. Even within the 311 dataset, potentially useful features like `closed_date_et` were excluded due to incomplete reporting; many requests lacked closure updates, limiting our ability to model resolution times or repair durations. As a result, our models are currently based solely on complaint data, and expanding the feature set with relevant infrastructure and traffic data would improve both the accuracy and practical utility of the predictions.

The framework currently uses fixed thresholds, such as identifying repeat complaints based on two prior reports within a 30-day window. While this design choice is practical and interpretable, it may not be the ideal definition of a repeat complaint to the DPW. Lastly, while the models have been validated, they have not yet been integrated into real-world DPW workflows. Future work should focus on deployment, monitoring model performance post-implementation, and ensuring outputs lead to meaningful action. A formal fairness audit including evaluating false positive and negative rates across demographic or geographic subgroups will also be essential to ensure that the potential benefits of our proposed models are distributed equitably across Pittsburgh.

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Appendix

GitHub Repository: https://github.com/pallavirajan00/90803_group17_finalproject

Data: <https://data.wprdc.org/dataset/311-data/resource/29462525-62a6-45bf-9b5e-ad2e1c06348d>